IEEE Envision'26

Assignment 1: Intro to Arduino
Date:28/03/2026

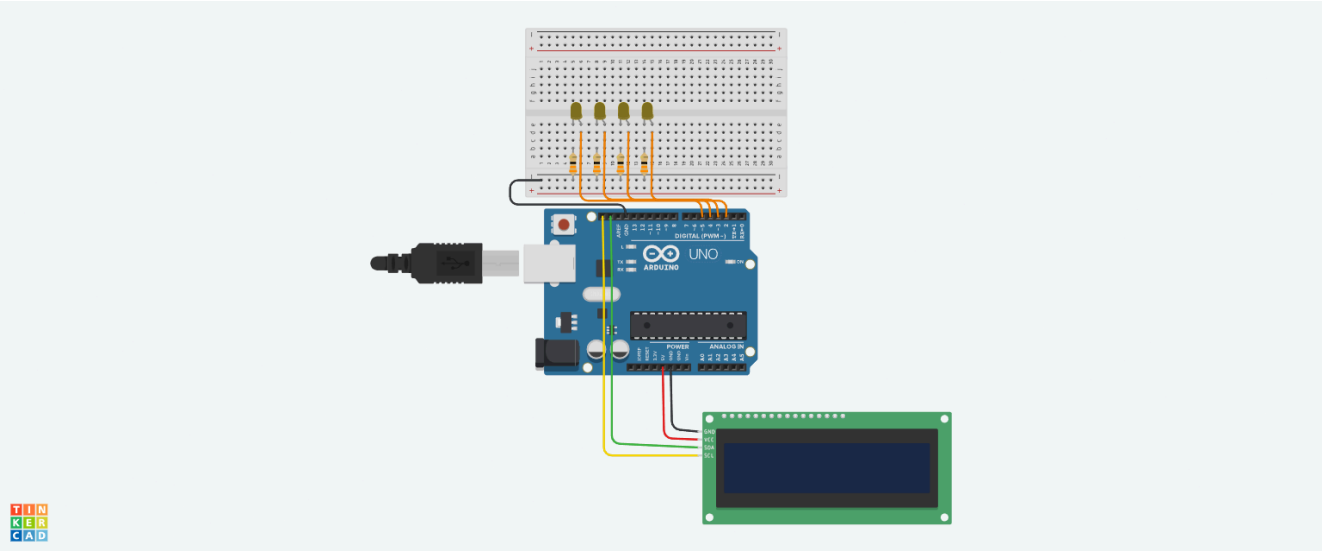
*** CIRCUITS MUST BE CREATED USING **TINKERCAD**

1. Create a circuit of your won choice using Arduino. Your circuit must contain of an **analog sensor**, an **output device (Digital or analog)** such as an LED, LCD Display, 7 Segment display or a buzzer. Submissions must **include screenshot of the circuit and code.**

2. Create a 4-bit binary adder using Arduino. The two inputs should be taken from user through Serial monitor as shown in the figure below

```
Enter A:
A = 1
Enter B:
B = 6
Sum = 7
```

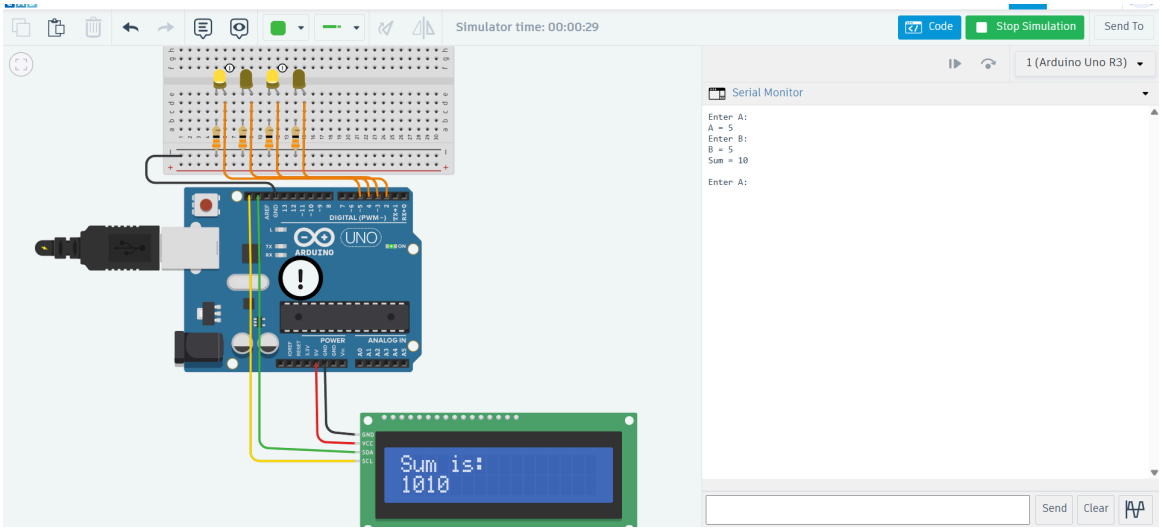
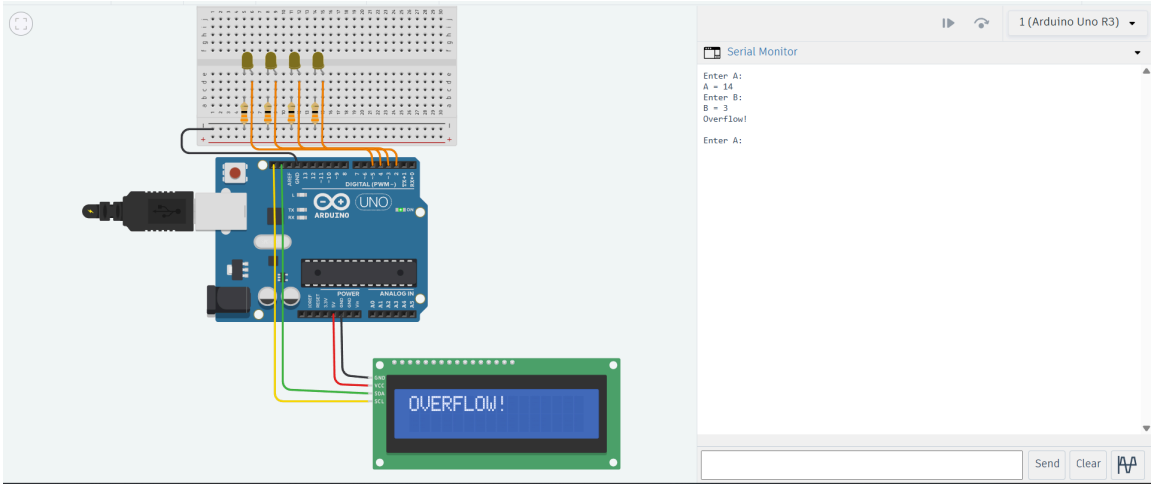
Input values are read from serial Monitor using ***Serial.parseInt()*** which takes input integer value the user types on his keyboard.Using an **I^2C Display (available as a pre Built circuit in Tinkercad),** you should also display the **binary value** of the final sum. In addition to that, your circuit must contain 4 LED's that display the binary value of sum **(GLOW IF THE BITVALUE=1 , OFF IF THE BITVALUE=0).Also if sum is greater than 15 (4 bit) , the I2C display must print "OVERFLOW".** The necessary Circuit diagrams, connections is provided and the correct test cases have also been Provided below.(Note: USE ONLY **I^2C DISPLAY** and USE **ADAFRUIT_12C_LCD** LIBRARY). Submission must contain **screenshot of circuit,** the **code** and **explanation of working of the code in brief.** (mainly how you are switching the LED's for that binary value)



CONNECTIONS:

LED'S (MSB-LEFT; LSB-RIGHT) — CONNECT TO DIGITAL PINS

I^2C DISPLAY– SDA TO A4/SDA PIN

SCL TO A5/SCL PIN

WORKING MODEL: